Statistics for Data Science II — Week 9 Notes

Topic: Parameter Estimation

> *These notes follow the standard IIT-M BS curriculum for Stats II Week 9. They cover definitions, derivations, formulas, and worked examples aligned with lectures L1–L10. They are topic-based notes, not a verbatim transcript.*

L1 — Statistical Problems in Real Life

Motivation

In probability, we *assume* a distribution (e.g., "this coin is fair, p = 0.5") and compute probabilities of events. In **statistics**, we do the reverse: we see data and try to figure out the distribution (or its parameters).

Typical real-life problems

- **Quality control**: What fraction of bulbs produced are defective?
- **Elections**: What fraction of voters support candidate A?
- **Medicine**: What is the mean blood pressure of adults?
- **Physics**: What is the true length of a table given noisy measurements?
- **ML**: Estimate weights of a model from training data.

Generic setup

- A **population** is described by a distribution $f_\theta(x)$ with unknown parameter θ.
- We observe a **sample** $X_1, X_2, \ldots, X_n$ (i.i.d. draws).
- Goal: use the sample to **estimate** θ.

Key idea

The unknown θ is a **fixed** quantity; the randomness comes from the sample. The function of the sample we use to estimate θ is itself a random variable — an **estimator**.

L2 — Introduction to Parameter Estimation

Definitions

- **Parameter** θ: an unknown constant describing the distribution (e.g., p for Bernoulli, μ and σ^2 for Normal, λ for Poisson).
- **Statistic**: any function of the sample $T(X_1, \ldots, X_n)$ that does not involve θ.
- **Estimator**: a statistic used to estimate θ. Denoted $\hat{\theta}$ or $\hat{\theta}_n$.
- **Estimate**: the numerical value of $\hat{\theta}$ for an observed sample.

> *Estimator = random variable. Estimate = realized number.*

Two kinds of estimates

- **Point estimate**: a single number (e.g., $\hat{p} = 0.47$).
- **Interval estimate (confidence interval)**: a range $[\hat{\theta}_L, \hat{\theta}_U]$ that contains θ with a stated probability.

Examples of estimators

Parameter	Natural estimator
p (Bernoulli)	sample proportion $\bar{X}$
μ (any distribution with finite mean)	sample mean $\bar{X} = \frac{1}{n}\sum X_i$
σ^2	sample variance $S^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$
λ (Poisson)	sample mean $\bar{X}$

L3 — Error in Estimation

Error

For an estimator $\hat{\theta}$ of parameter θ:

$$\text{error} = \hat{\theta} - \theta$$

This is a random variable (depends on the sample).

Mean Squared Error (MSE)

$$\text{MSE}(\hat{\theta}) = \mathbb{E}\left[(\hat{\theta} - \theta)^2\right]$$

MSE captures average squared deviation. Smaller is better.

Decomposition

$$\text{MSE}(\hat\theta) = \underbrace{\text{Var}(\hat\theta)}{\text{spread}} + \underbrace{\bigl(\mathbb{E}[\hat\theta] - \theta\bigr)^2}{\text{bias}^2}$$

This is the **bias–variance decomposition** — the cornerstone identity in estimation (and later, ML).

Why it matters

- Low bias + low variance → low MSE → great estimator.
- An estimator can be unbiased but high-variance (useless) or biased but low-variance (sometimes preferable).

L4 — Bias, Variance & Risk of an Estimator

Bias

$$\text{Bias}(\hat\theta) = \mathbb{E}[\hat\theta] - \theta$$

- **Unbiased** if Bias = 0, i.e., $\mathbb{E}[\hat\theta] = \theta$.
- Example: $\bar{X}$ is unbiased for μ because $\mathbb{E}[\bar{X}] = \mu$.
- Example: $S^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$ is unbiased for σ^2. The divisor $n-1$ (not n) is **precisely** what removes the bias.

Variance

$$\text{Var}(\hat\theta) = \mathbb{E}\left[(\hat\theta - \mathbb{E}[\hat\theta])^2\right]$$

For the sample mean: $\text{Var}(\bar{X}) = \sigma^2/n$. Variance shrinks as sample size grows.

Risk

For a loss function $L(\hat\theta, \theta)$, the **risk** is

$$R(\hat\theta, \theta) = \mathbb{E}[L(\hat\theta, \theta)]$$

With squared-error loss $L(\hat\theta, \theta) = (\hat\theta - \theta)^2$, risk = MSE.

Comparing estimators

$\hat\theta_1$ **dominates** $\hat\theta_2$ if $R(\hat\theta_1, \theta) \le R(\hat\theta_2, \theta)$ for all θ, with strict inequality somewhere.

L5 — Method of Moments (MoM / MME)

Idea

Match **population moments** to **sample moments**, solve for the parameter.

- k-th population moment: $\mu_k = \mathbb{E}[X^k]$ (a function of θ).
- k-th sample moment: $m_k = \frac{1}{n}\sum_{i=1}^{n} X_i^k$.

Set $\mu_k(\theta) = m_k$ and solve.

Recipe

1. Count parameters. If there are p parameters, use moments up through μ_p.
2. Write population moments as functions of the parameters.
3. Replace each μ_k with the corresponding sample moment m_k.
4. Solve the system for $\hat{\theta}_{\text{MME}}$.

Example 1 — Bernoulli(p)

- $\mu_1 = p$.
- Set $p = \bar{X} \implies \hat{p}_{\text{MME}} = \bar{X}$.

Example 2 — Poisson(λ)

- $\mu_1 = \lambda$. So $\hat{\lambda}_{\text{MME}} = \bar{X}$.

Example 3 — Normal(μ, σ^2)

Two parameters, so use μ_1 and μ_2:

- $\mu_1 = \mu \implies \hat{\mu} = \bar{X}$.
- $\mu_2 = \mu^2 + \sigma^2 \implies \hat{\sigma}^2 = m_2 - \bar{X}^2 = \frac{1}{n}\sum(X_i - \bar{X})^2$.

Note: MoM gives divisor n, not $n-1$, so MoM variance is **biased**.

Example 4 — Uniform$(0, \theta)$

- $\mu_1 = \theta/2 \implies \hat{\theta}_{\text{MME}} = 2\bar{X}$.

Pros / cons

- Pros: simple, always works if moments exist.
- Cons: not always efficient; can give silly estimates (e.g., Uniform above can produce $\hat{\theta} < \max X_i$, which is impossible).

L6 — Maximum Likelihood Estimation (MLE)

Likelihood function

Given observed data $x_1, \ldots, x_n$ and parameter θ:

$$L(\theta) = \prod_{i=1}^{n} f_\theta(x_i)$$

(product of the PMF/PDF). The likelihood treats the data as fixed and θ as the variable.

Log-likelihood

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^{n} \log f_\theta(x_i)$$

Easier to work with (sums instead of products).

MLE

$$\hat{\theta}_{\text{MLE}} = \arg\max_\theta L(\theta) = \arg\max_\theta \ell(\theta)$$

Recipe

1. Write $L(\theta)$ or preferably $\ell(\theta)$.
2. Differentiate: $\frac{d\ell}{d\theta}$.
3. Set to 0, solve for θ.
4. Verify it's a maximum (second derivative negative, or check boundary).

Example 1 — Bernoulli(p)

Data: $x_1, \ldots, x_n \in 0, 1$, let $s = \sum x_i$.

- $L(p) = p^s (1-p)^{n-s}$.
- $\ell(p) = s \log p + (n-s) \log(1-p)$.
- $\ell'(p) = s/p - (n-s)/(1-p) = 0 \implies \hat{p}_{\text{MLE}} = s/n = \bar{X}$.

Example 2 — Poisson(λ)

- $\ell(\lambda) = -n\lambda + (\sum x_i) \log \lambda - \sum \log(x_i!)$.
- $\ell'(\lambda) = -n + \sum x_i / \lambda = 0 \implies \hat{\lambda}_{\text{MLE}} = \bar{X}$.

Example 3 — Normal(μ, σ^2)

- $\hat{\mu}_{\text{MLE}} = \bar{X}$.
- $\hat{\sigma}^2_{\text{MLE}} = \frac{1}{n} \sum (X_i - \bar{X})^2$ (biased; divisor n).

Example 4 — Uniform$(0, \theta)$

- $L(\theta) = \theta^{-n}$ for $\theta \geq \max X_i$, else 0.
- L is decreasing in θ, so MLE = smallest allowed value.
- $\hat{\theta}_{\text{MLE}} = \max_i X_i$.

(Compare with MoM: $2\bar{X}$. MLE is much better here.)

L7 — Evaluation of ML Estimators

Key properties of MLEs (asymptotic, under regularity)

1. **Consistency**: $\hat{\theta}_{\text{MLE}} \xrightarrow{P} \theta$ as $n \to \infty$.
2. **Asymptotic normality**: $\sqrt{n}(\hat{\theta}_{\text{MLE}} - \theta) \xrightarrow{d} \mathcal{N}(0, I(\theta)^{-1})$ where $I(\theta)$ is the **Fisher information**.
3. **Asymptotic efficiency**: achieves the Cramér–Rao lower bound asymptotically.
4. **Invariance**: if $\hat{\theta}$ is the MLE of θ, then $g(\hat{\theta})$ is the MLE of $g(\theta)$ for any function g.

Fisher information (intuition)

$$I(\theta) = -\mathbb{E}\left[\frac{d^2\ell}{d\theta^2}\right]$$

Measures how "curved" the log-likelihood is at θ. Higher curvature ⇒ more informative data ⇒ smaller variance for MLE.

Cramér–Rao Lower Bound (CRLB)

For any unbiased estimator $\hat{\theta}$:

$$\text{Var}(\hat{\theta}) \geq \frac{1}{n, I(\theta)}$$

No unbiased estimator can do better than this. An estimator hitting this bound is called **efficient**.

Small-sample caveats

MLEs are not always unbiased. Example: $\hat{\sigma}^2_{\text{MLE}}$ has bias $-\sigma^2/n$. But as $n \to \infty$, bias $\to 0$.

L8 — Finding MME & ML Estimators (Problems)

Problem 1 — Geometric(p)

- PMF: $P(X = k) = (1-p)^{k-1}p$ for $k = 1, 2, \ldots$; $\mathbb{E}[X] = 1/p$.
- **MoM**: $\bar{X} = 1/p \implies \hat{p}_{\text{MME}} = 1/\bar{X}$.
- **MLE**: $\ell(p) = n \log p + (\sum x_i - n) \log(1-p)$. Solve: $\hat{p}_{\text{MLE}} = 1/\bar{X}$. (Same as MoM here.)

Problem 2 — Exponential(λ)

- $f(x) = \lambda e^{-\lambda x}$, $\mathbb{E}[X] = 1/\lambda$.
- **MoM**: $\hat{\lambda}_{\text{MME}} = 1/\bar{X}$.
- **MLE**: $\ell(\lambda) = n \log \lambda - \lambda \sum x_i \implies \hat{\lambda}_{\text{MLE}} = 1/\bar{X}$.

Problem 3 — Uniform(a, b)

- $\mu_1 = (a+b)/2$, $\sigma^2 = (b-a)^2/12$.
- **MoM**: solve two equations $\to \hat{a} = \bar{X} - \sqrt{3}, s$, $\hat{b} = \bar{X} + \sqrt{3}, s$, where $s^2 = m_2 - \bar{X}^2$.
- **MLE**: $\hat{a}_{\text{MLE}} = \min X_i$, $\hat{b}_{\text{MLE}} = \max X_i$.

Problem 4 — Normal (both μ, σ^2 unknown)

Both methods give $\hat{\mu} = \bar{X}$. For variance:

- **MoM / MLE**: divisor n (biased).
- **Unbiased estimator** S^2: divisor $n-1$.

General tips

- Always check whether the likelihood is maximised at an **interior** point (calculus) or at a **boundary** (like the uniform max).
- Log-likelihood usually has better algebra.
- When likelihood is monotone in θ, the MLE sits at an endpoint of the feasible set.

L9 — Properties of Estimators

Unbiasedness

$\mathbb{E}[\hat{\theta}] = \theta$. Desirable but not sufficient.

Consistency

$\hat{\theta}_n \xrightarrow{P} \theta$: for large n, the estimator concentrates around θ.

Sufficient condition: Bias $\to 0$ **and** Variance $\to 0$ as $n \to \infty \Rightarrow$ MSE $\to 0 \Rightarrow$ consistent.

Efficiency

Among all unbiased estimators, the **efficient** one has minimum variance. If it hits the CRLB, it's the **UMVUE** (Uniformly Minimum Variance Unbiased Estimator).

Sufficiency

A statistic T is **sufficient** for θ if the conditional distribution of the data given T does not depend on θ. Intuition: T captures all information about θ.

- Example: $\sum X_i$ is sufficient for λ in Poisson.
- Example: $(\sum X_i, \sum X_i^2)$ is sufficient for (μ, σ^2) in Normal.

Hierarchy (good-to-great)

Unbiased $\longrightarrow$ Consistent $\longrightarrow$ Efficient $\longrightarrow$ Sufficient-based (Rao–Blackwell improvement) $\longrightarrow$ UMVUE.

Rao–Blackwell theorem (stated, not derived)

If $\hat{\theta}$ is unbiased and T is sufficient, then $\mathbb{E}[\hat{\theta} \mid T]$ is unbiased and has variance no larger than $\hat{\theta}$.

L10 — Confidence Intervals

Point vs interval

A point estimate gives no sense of uncertainty. A **confidence interval (CI)** reports a range.

Definition

A $100(1-\alpha)$ CI for θ is a random interval $[L(X), U(X)]$ such that

$$P(L(X) \leq \theta \leq U(X)) = 1 - \alpha$$

Typical choice: $\alpha = 0.05 \Rightarrow 95\%$ CI.

Interpretation (careful!)

"If we repeated the experiment many times, 95% of the constructed intervals would contain the true θ."
Not: "there is a 95% probability that θ is inside this particular interval" (once we observe the data, θ is either in or out).

CI for the mean μ — known σ (Normal data)

$$\bar{X} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

where $z_{\alpha/2}$ is the upper-$\alpha/2$ quantile of $\mathcal{N}(0,1)$. For 95%: $z_{0.025} = 1.96$.

CI for the mean μ — unknown σ (Normal data)

$$\bar{X} \pm t_{\alpha/2,n-1} \cdot \frac{S}{\sqrt{n}}$$

t-distribution with $n-1$ degrees of freedom. For large n, $t \to z$.

CI for a proportion p (large n)

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Relationship to hypothesis testing (preview of Week 10+)

A 95% CI for θ contains exactly those values that would **not be rejected** in a two-sided test at significance level 5%.

Sample size planning

Want margin of error $\leq E$ with 95% confidence?

$$n \geq \left(\frac{z_{\alpha/2}, \sigma}{E}\right)^2$$

Quick Formula Sheet

Quantity	Formula
Sample mean	$\bar{X} = \frac{1}{n}\sum X_i$
Sample variance (unbiased)	$S^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$
Bias	$\mathbb{E}[\hat{\theta}] - \theta$
MSE	$\mathrm{Var}(\hat{\theta}) + \mathrm{Bias}^2$
Likelihood	$L(\theta) = \prod f_\theta(x_i)$
Log-likelihood	$\ell(\theta) = \sum \log f_\theta(x_i)$
MLE condition	$\ell'(\theta) = 0$
Fisher info	$I(\theta) = -\mathbb{E}[\ell''(\theta)]$

Quantity	Formula
CRLB	$\mathrm{Var}(\hat{\theta}) \geq \frac{1}{nI(\theta)}$
95% CI (known σ)	$\bar{X} \pm 1.96, \sigma/\sqrt{n}$
95% CI (unknown σ)	$\bar{X} \pm t_{0.025,n-1}, S/\sqrt{n}$

Common MME vs MLE Comparison

Distribution	MME	MLE
Bernoulli(p)	$\bar{X}$	$\bar{X}$
Poisson(λ)	$\bar{X}$	$\bar{X}$
Exp(λ)	$1/\bar{X}$	$1/\bar{X}$
Geom(p)	$1/\bar{X}$	$1/\bar{X}$
Normal(μ)	$\bar{X}$	$\bar{X}$
Normal(σ^2)	$\frac{1}{n}\sum(X_i - \bar{X})^2$	$\frac{1}{n}\sum(X_i - \bar{X})^2$
Uniform$(0, \theta)$	$2\bar{X}$	$\max X_i$

Exam-Style Practice Problems

P1. Let $X_1, \ldots, X_n \sim \mathrm{Exp}(\lambda)$. Find MoM and MLE of λ. Are they unbiased?

P2. Let $X_1, \ldots, X_n \sim \mathcal{N}(\mu, 1)$. Show that $\bar{X}$ is the MLE of μ, is unbiased, and achieves the CRLB.

P3. Let $X_1, \ldots, X_n \sim \mathrm{Uniform}(0, \theta)$. Compute bias and variance of (a) $2\bar{X}$ and (b) $\max X_i$. Which has smaller MSE?

P4. A sample of 100 has mean 50 and $S = 10$. Construct a 95% CI for μ.

P5. For Poisson(λ), show that $\bar{X}$ is unbiased, consistent, and efficient.

End of Week 9 notes. Revisit lecture videos for spoken intuition and chalkboard derivations.